

Bijay Pandey

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PROFESSIONAL SUMMARY

Master of Applied Science student in Software Systems Engineering at the University of Regina with over five years of professional experience in software engineering, cloud infrastructure, and Site Reliability Engineering. Current graduate research focuses on Artificial Intelligence and Machine Learning for data-driven physical systems. Hands-on experience with Python, SQL, Pandas, NumPy, Scikit-learn, statistical and predictive modelling, model evaluation, AWS, and large-scale production systems. Strong analytical background in mathematics, numerical methods, data analysis, and algorithms, with experience translating technical findings into reproducible code, documentation, and clear communication.

EDUCATION

Master of Applied Science - Software Systems Engineering
University of Regina, Regina, SK

Jan 2026 - Present
Expected completion Apr 2028

Bachelor of Electronics and Communication Engineering
Tribhuvan University, Lalitpur, Nepal

Nov 2015 - Sep 2019

- **Honours Thesis:** ECC Based Secured Telemetry of Mobile Weather Proxies
- **Relevant Coursework:** Artificial Intelligence/Machine Learning, Data Mining, Numerical Methods, Mathematics (Calculus, Linear Algebra, Probability and Statistics), Algorithms, Object-Oriented Programming, Computer Networks

RESEARCH & ANALYTICAL INTERESTS

- **Machine Learning & Predictive Modelling**
- **Physics-Informed / Explainable AI**
- **Statistical Model Evaluation**
- **Exploratory Data Analysis & Data Quality**
- **Predictive Maintenance & Reliability**
- **Cloud-based Data & ML Systems**

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, Pandas, NumPy, Scikit-learn, Matplotlib, MLflow, Bash
- **Machine Learning:** Exploratory Data Analysis, Data Preprocessing, Feature Engineering, Regression, Classification, Cross-Validation, Model Evaluation, Explainability (SHAP/LIME)
- **Mathematics & Analytics:** Probability and Statistics, Linear Algebra, Calculus, Numerical Methods, Algorithms, Analytical Problem Solving
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
- **Cloud & DevOps:** AWS, GCP, CloudWatch, Docker, Kubernetes, Terraform, CloudFormation, CI/CD
- **Software & Systems:** Django, Django REST Framework, REST APIs, Git, GitHub, Linux, Distributed Systems
- **Monitoring & Observability:** Prometheus, Grafana, CloudWatch, New Relic

RESEARCH & TEACHING EXPERIENCE

Graduate Researcher / Teaching Assistant / Lab Instructor
University of Regina, Regina, SK

2026 - Present

- Conduct graduate research in AI/ML for physical systems, focusing on predictive maintenance, fault analysis, model evaluation, and reliable decision support.
- Develop reproducible Python-based workflows for data analysis, feature engineering, validation, and interpretability while reviewing literature and documenting research findings.
- Support laboratory instruction by explaining technical concepts, guiding students through exercises, and providing structured feedback.

PROFESSIONAL EXPERIENCE

Site Reliability Engineer
CloudFactory, Lalitpur, Nepal

May 2022 - Dec 2025

- Designed, deployed, and maintained scalable cloud-based applications and production infrastructure, ensuring high availability, reliability, and performance.
- Developed Python automation for large-scale distributed production systems, reducing manual operational work and improving repeatability of support workflows.
- Analyzed logs, metrics, and system behaviour using CloudWatch, Prometheus, Grafana, and New Relic to identify anomalies, troubleshoot incidents, and investigate performance bottlenecks.
- Automated infrastructure provisioning and operational workflows using Terraform, AWS CloudFormation, Bash, and Infrastructure-as-Code practices.
- Built and maintained CI/CD pipelines and supported containerized applications with Docker and Kubernetes across cloud environments.
- Performed root-cause analysis for production issues and documented findings, corrective actions, runbooks, and operational procedures for reliable incident response.

- Collaborated with software engineering and operations teams to improve production releases, system performance, and service reliability.

Software Engineer - Backend Development & DevOps

Feb 2020 - Apr 2022

Leapfrog Technology, Kathmandu, Nepal

- Designed, developed, and maintained enterprise backend applications using Django and Django REST Framework, delivering scalable and secure software services.
- Designed and implemented REST APIs, backend services, business logic, and system integrations for data-driven enterprise applications.
- Developed and optimized relational database schemas and SQL queries using PostgreSQL/MySQL, improving data integrity, performance, and scalability.
- Collaborated with cross-functional teams to design, implement, test, debug, and deploy software solutions in production environments.
- Built CI/CD workflows and supported Docker/Kubernetes deployments while contributing to technical documentation and knowledge-sharing sessions.

SELECTED RESEARCH & ENGINEERING PROJECTS

Physics-Informed AI for Predictive Maintenance & Battery Health Analysis

MASc Research - Ongoing

- Researching physics-informed and data-driven AI for predictive maintenance, with emphasis on lithium-ion battery fault diagnosis, degradation modelling, and remaining useful life prediction.
- Building Python-based workflows for data preprocessing, feature engineering, predictive modelling, validation, and explainability using Pandas, NumPy, Scikit-learn, and SHAP/LIME.
- Investigating robust and interpretable decision support by combining physical-system knowledge with machine-learning methods under varying operating conditions.

ECC Based Secured Telemetry of Mobile Weather Proxies

Tribhuvan University

- Designed and evaluated a secure IoT telemetry framework using ECC and RSA, comparing security, latency, and computational efficiency under resource constraints.
- Built a real-time monitoring dashboard for encrypted telemetry data using Python and embedded/IoT technologies.

LEADERSHIP & TECHNICAL COMMUNICATION

LOCUS Exhibition - Organizer, Volunteer and Hardware Fellowship Mentor

- Mentored undergraduate students on embedded systems, IoT development, programming, and hands-on technical projects.
- Organized technical workshops and competitive programming activities and guided participants on algorithms, problem solving, and coding practices.